Spectral Risk Sharing between Coupled Economies

A Unifying Treatise on DFT Compression, Schur-Complement Hedging, and Mutual Risk Reduction

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Abstract

We study two economies—the *original* and the *copycat*—whose states are n-dimensional random vectors sharing a common probability space. The copycat economy is formed by applying the unitary Discrete Fourier Transform (DFT) to the original state vector and retaining only the $k < n$ lowest-frequency spectral components. Under the standard assumption that the state covariance is positive definite, we establish three main results. *First* (Theorem 5.4), the copycat has strictly lower total-variance risk than the original; the deficit equals the trace of the covariance projected onto the discarded high-frequency subspace. *Second* (Theorem 5.5), broadcasting the compressed state allows the original economy to adopt it and achieve the same risk as the copycat—a pure compression-and-adoption gain. *Third*, and most substantially (Theorem 5.7), when the original broadcasts its full state to the copycat, the copycat can form a minimum-variance linear combination of its existing low-frequency state and the newly received high-frequency complement; the optimal hedge coefficient equals the cross-spectral Wiener gain, and the minimum attainable risk is the Schur complement $\mathrm{tr}(\mathbf{\Sigma}/\mathbf{\Sigma}_{QQ})$, which lies strictly below the pre-communication level whenever low- and high-frequency components are cross-correlated. These three results combine into a *Mutual Communication Corollary*: under the bilateral exchange of state variables the aggregate risk of *both* economies decreases strictly, establishing that spectral compression and cross-economy communication are unconditionally mutually beneficial. We situate the framework within the broader literature on principal component analysis, Kalman filtering, federated learning, and financial risk decomposition, and supply a formal algorithm together with a numerical illustration.

The treatise ends with "The End"

Keywords: Discrete Fourier Transform, spectral risk sharing, Schur complement, spectral hedging, covariance projection, mutual communication, PCA, Wiener filter.

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1 Introduction

A central question in mathematical economics, signal processing, and machine learning is whether two interacting agents can reduce their individual exposure to uncertainty simply by exchanging information about their state variables—*without* any explicit transfer of wealth or physical resources.

The classical answer from portfolio theory [11] is affirmative: agents can pool risks through coordinated asset positions. A more recent answer from machine learning is that agents benefit from sharing compressed representations of their data, a setting formalised by federated learning [12] and by the information-theoretic framework of the information bottleneck [16].

The present treatise provides a third, frequency-domain answer rooted in spectral analysis. We posit that the *original economy* $\mathcal{O}$ holds a zero-mean state vector $\mathbf{x} \in \mathbb{R}^n$ with positive-definite covariance $\mathbf{\Sigma}$. The *copycat economy* $\mathcal{C}$ does not observe $\mathbf{x}$ directly; it forms its state by applying the unitary DFT matrix $\mathbf{F}$ to $\mathbf{x}$ and projecting onto the subspace spanned by the $k < n$ lowest-frequency modes. The resulting state $\mathbf{y} = P_k\mathbf{x}$ is the band-limited spectral portrait of $\mathbf{x}$, and its total variance is strictly smaller than that of $\mathbf{x}$ whenever $\mathbf{\Sigma}$ is positive definite.

The treatise makes four principal contributions.

(i) **Risk ordering.** We prove that $\mathbf{y}$ has strictly lower total variance than $\mathbf{x}$, with the gap equal to $\operatorname{tr}(\mathbf{\Sigma}Q_k)$—the variance carried by the discarded high-frequency modes.

(ii) **Original benefits.** We show that when $\mathcal{C}$ broadcasts $\mathbf{y}$ to $\mathcal{O}$, the original economy can adopt it as its new state and eliminate the same high-frequency variance gap, without any knowledge of the DFT structure of $\mathbf{\Sigma}$.

(iii) **Copycat benefits.** We prove that when $\mathcal{O}$ broadcasts its full state $\mathbf{x}$ to $\mathcal{C}$, the copycat can construct a hedged state $\mathbf{y}^* = P_k\mathbf{x} + B^*Q_k\mathbf{x}$ whose risk equals the Schur complement $\operatorname{tr}(\mathbf{\Sigma}/\mathbf{\Sigma}_{QQ})$, which is strictly smaller than $\operatorname{tr}(\mathbf{\Sigma}_{PP})$ whenever the cross-spectral covariance block $\mathbf{\Sigma}_{PQ}$ is nonzero. The optimal hedge gain $B^* = -\mathbf{\Sigma}_{PQ}\mathbf{\Sigma}_{QQ}^{-1}$ is the (negative) Wiener filter of the high-frequency signal on the low-frequency state.

(iv) **Mutual gain.** We combine items (i)–(iii) to prove that bilateral communication yields a total risk reduction of $\operatorname{tr}(\mathbf{\Sigma}Q_k) + \operatorname{tr}(\mathbf{\Sigma}_{PQ}\mathbf{\Sigma}_{QQ}^{-1}\mathbf{\Sigma}_{QP}) > 0$, which is strictly positive under the blanket assumption $\mathbf{\Sigma} \succ 0$.

Treatise organisation. Section 2 fixes notation and recalls key facts from matrix analysis and the DFT. Section 3 introduces the two-economy model and the risk measure. Section 4 develops the spectral projection calculus. Section 5 states and proves the main results. Section 6 presents the mutual communication algorithm. Section 7 connects the framework to PCA, federated learning, Kalman filtering, financial risk, and spectral graph theory. Section 8 provides a fully worked numerical example. Section 9 discusses extensions and limitations. Section 10 concludes. A glossary and bibliography follow.

2 Notation and Preliminaries

2.1 Linear algebra

Boldface lowercase letters denote column vectors ($\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$); boldface uppercase letters denote matrices ($\mathbf{F} \in \mathbb{C}^{n\times n}$, $\mathbf{\Sigma} \in \mathbb{R}^{n\times n}$). $(\cdot)^\top$ denotes the real transpose; $(\cdot)^H$ the conjugate transpose. $\mathbf{I}_n$ is the $n \times n$ identity matrix (or simply $\mathbf{I}$ when dimension is clear). $\operatorname{diag}(d_1, \ldots, d_n)$ is the diagonal matrix with entries d_i.

For a symmetric matrix A, write $A \succ 0$ ($A \succeq 0$) if A is positive definite (positive semidefinite). The *trace* satisfies $\operatorname{tr}(A + B) = \operatorname{tr}(A) + \operatorname{tr}(B)$ and the cyclicity identity $\operatorname{tr}(AB) = \operatorname{tr}(BA)$ for conformable A, B. If $A \succeq 0$ and $B \succeq 0$ then $\operatorname{tr}(AB) \geq 0$.

Definition 2.1 (Schur complement)**.** Let $M = \left(\begin{smallmatrix} A & B \\ C & D \end{smallmatrix}\right)$ with D invertible. The *Schur complement of D in M* is

$$M/D \;:=\; A - BD^{-1}C.$$

If $M \succ 0$ then $D \succ 0$, $M/D \succ 0$, and $\det M = \det D \cdot \det(M/D)$.

2.2 The Discrete Fourier Transform

Definition 2.2 (DFT matrix)**.** For $n \geq 1$, let $\omega_n = e^{-2\pi \mathrm{i}/n}$. The *DFT matrix* $\mathbf{F} \in \mathbb{C}^{n\times n}$ has (j,k)-entry

$$[\mathbf{F}]_{jk} \;=\; \frac{1}{\sqrt{n}}\,\omega_n^{jk}, \qquad 0 \leq j,k \leq n-1.$$

$\mathbf{F}$ is unitary: $\mathbf{F}\mathbf{F}^H = \mathbf{F}^H\mathbf{F} = \mathbf{I}_n$.

Lemma 2.3 (Parseval's identity)**.** *For any* $\mathbf{x} \in \mathbb{C}^n$, $\|\mathbf{F}\mathbf{x}\|^2 = \|\mathbf{x}\|^2$. *More generally, for any* $\mathbf{\Sigma} \succeq 0$, $\mathrm{tr}(\mathbf{F}\mathbf{\Sigma}\mathbf{F}^H) = \mathrm{tr}(\mathbf{\Sigma})$.

Proof. $\|\mathbf{F}\mathbf{x}\|^2 = (\mathbf{F}\mathbf{x})^H(\mathbf{F}\mathbf{x}) = \mathbf{x}^H\mathbf{F}^H\mathbf{F}\mathbf{x} = \|\mathbf{x}\|^2$. The trace identity follows by cyclicity: $\mathrm{tr}(\mathbf{F}\mathbf{\Sigma}\mathbf{F}^H) = \mathrm{tr}(\mathbf{F}^H\mathbf{F}\,\mathbf{\Sigma}) = \mathrm{tr}(\mathbf{\Sigma})$. □

Lemma 2.3 guarantees that the DFT is an *isometry*: it neither creates nor destroys variance. Risk differences therefore arise entirely from the choice of which spectral modes to retain, not from the transform itself.

2.3 Orthogonal projections

A matrix $P \in \mathbb{R}^{n\times n}$ is an *orthogonal projection* if $P^2 = P$ (idempotent) and $P^\top = P$ (symmetric). Every orthogonal projection satisfies $0 \preceq P \preceq \mathbf{I}$ and $\mathrm{rank}(P) = \mathrm{tr}(P)$. If P is an orthogonal projection then so is $Q := \mathbf{I} - P$, and $PQ = QP = 0$.

3 The Two-Economy Framework

3.1 State spaces and covariance

Definition 3.1 (Original economy $\mathcal{O}$)**.** Economy $\mathcal{O}$ is described by a zero-mean random vector $\mathbf{x} \in \mathbb{R}^n$ with positive-definite covariance $\mathbf{\Sigma} = \mathrm{Cov}(\mathbf{x}) = \mathbb{E}[\mathbf{x}\mathbf{x}^\top] \succ 0$.

The assumption $\mathbf{\Sigma} \succ 0$ is both standard and substantive: it rules out exact linear dependences among the n state variables, ensuring that every DFT mode carries strictly positive variance.

Definition 3.2 (Copycat economy $\mathcal{C}$)**.** Fix an integer $1 \leq k < n$. Economy $\mathcal{C}$ holds state $\mathbf{y} = P_k\mathbf{x}$, where

$$P_k \;:=\; \mathbf{F}^H D_k \mathbf{F}, \qquad D_k \;:=\; \mathrm{diag}\big(\underbrace{1,\ldots,1}_{k},\,\underbrace{0,\ldots,0}_{n-k}\big), \tag{1}$$

with $\mathbf{F}$ as in Definition 2.2.

P_k is the orthogonal projection onto the k-dimensional *low-frequency subspace* spanned by the first k columns of $\mathbf{F}^H$. Its complement is

$$Q_k := \mathbf{I} - P_k = \mathbf{F}^H(\mathbf{I} - D_k)\mathbf{F},$$

and the partition $\mathbf{I} = P_k + Q_k$ is orthogonal: $P_kQ_k = Q_kP_k = 0$.

3.2 The risk measure

Definition 3.3 (Total variance risk)**.** The *risk* of a zero-mean random vector $\mathbf{v} \in \mathbb{R}^n$ is

$$\mathcal{R}(\mathbf{v}) \;:=\; \mathrm{tr}\big(\mathrm{Cov}(\mathbf{v})\big) \;=\; \sum_{j=1}^{n} \mathrm{Var}(v_j).$$

By extension, $\mathcal{R}(\mathcal{O}) := \mathcal{R}(\mathbf{x})$ and $\mathcal{R}(\mathcal{C}) := \mathcal{R}(\mathbf{y})$.

Total variance is the sum of individual variances and equals the squared Frobenius norm of the Cholesky factor of the covariance matrix. It is the objective minimised in minimum-variance portfolio construction [11], the loss reconstructed by PCA [8], and the integrated mean-square prediction error in Gaussian process regression [14]. Alternative risk measures are discussed in Section 9.

4 Spectral Projection Calculus

4.1 Covariance block decomposition

Apply the orthogonal partition $\mathbf{I} = P_k + Q_k$ to $\mathbf{\Sigma}$:

$$\mathbf{\Sigma}_{PP} := P_k \mathbf{\Sigma} P_k, \tag{2}$$

$$\mathbf{\Sigma}_{QQ} := Q_k \mathbf{\Sigma} Q_k, \tag{3}$$

$$\mathbf{\Sigma}_{PQ} := P_k \mathbf{\Sigma} Q_k = \mathbf{\Sigma}_{QP}^\top. \tag{4}$$

Since $\mathbf{\Sigma} \succ 0$ and both P_k and Q_k have rank at least one (for $1 \le k < n$), we have $\mathbf{\Sigma}_{PP} \succ 0$ and $\mathbf{\Sigma}_{QQ} \succ 0$ on their respective ranges.

The trace additivity $\mathrm{tr}(P_k Q_k) = 0$ and cyclicity give

$$\mathrm{tr}(\mathbf{\Sigma}) = \mathrm{tr}(\mathbf{\Sigma} P_k) + \mathrm{tr}(\mathbf{\Sigma} Q_k) = \mathrm{tr}(\mathbf{\Sigma}_{PP}) + \mathrm{tr}(\mathbf{\Sigma}_{QQ}), \tag{5}$$

confirming that the cross-blocks $\mathbf{\Sigma}_{PQ}, \mathbf{\Sigma}_{QP}$ make no contribution to the total risk.

4.2 Risk of the copycat state

Lemma 4.1 (Risk of the projected state)**.** $\mathcal{R}(\mathbf{y}) = \mathrm{tr}(\mathbf{\Sigma}_{PP}) = \mathrm{tr}(\mathbf{\Sigma} P_k)$.

Proof. $\mathrm{Cov}(\mathbf{y}) = \mathrm{Cov}(P_k \mathbf{x}) = P_k \mathbf{\Sigma} P_k = \mathbf{\Sigma}_{PP}$, and $\mathrm{tr}(\mathbf{\Sigma}_{PP}) = \mathrm{tr}(P_k \mathbf{\Sigma} P_k) = \mathrm{tr}(\mathbf{\Sigma} P_k^2) = \mathrm{tr}(\mathbf{\Sigma} P_k)$ by cyclicity and idempotency. □

4.3 Cross-spectral correlation

The off-diagonal block $\mathbf{\Sigma}_{PQ} = P_k \mathbf{\Sigma} Q_k$ measures linear covariance between the retained modes $\{0, \ldots, k-1\}$ and the discarded modes $\{k, \ldots, n-1\}$. When $\mathbf{x}$ is *wide-sense stationary*—i.e. $\mathbf{\Sigma}$ is a circulant matrix—the DFT diagonalises $\mathbf{\Sigma}$ exactly (Wiener–Khinchin theorem) and $\mathbf{\Sigma}_{PQ} = 0$. Cross-spectral correlation is therefore a hallmark of non-stationarity, regime-switching, or nonlinear inter-frequency coupling, all of which are empirically prevalent in economic time series.

Figure 1 illustrates the overall architecture: the forward path (solid arrows) constructs the copycat state via DFT compression; the feedback paths (dashed arrows) implement the bilateral communication studied in Section 5.

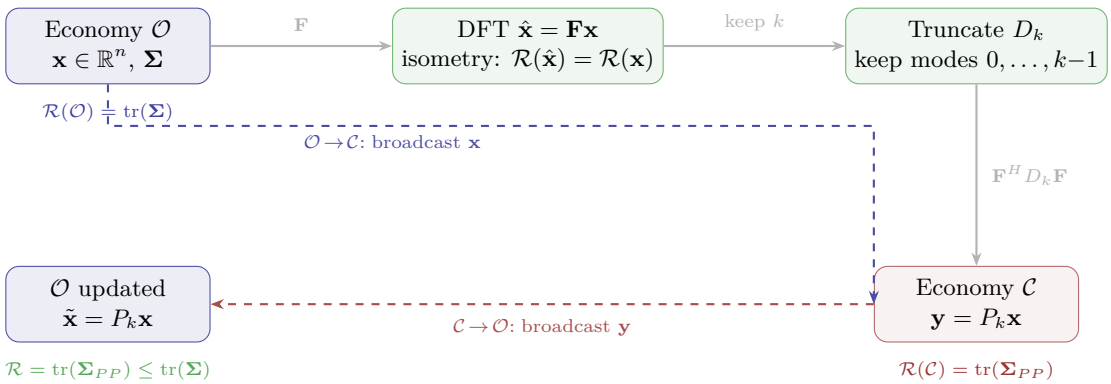

Figure 1: Architecture of the two-economy framework.

The forward path (solid arrows) constructs the copycat state $\mathbf{y}$ by DFT compression of $\mathbf{x}$. Dashed arrows represent the bilateral communication channel studied in Theorems 5.5 and 5.7.

5 Main Results

We first state the two conditions under which the risk reductions are strict, then prove the three theorems and the corollary.

Condition 5.1 (C1: high-frequency modes carry variance)**.** $\mathrm{tr}(\mathbf{\Sigma} Q_k) > 0$.

Remark 5.2. Under $\mathbf{\Sigma} \succ 0$ and $1 \le k < n$, Condition 5.1 holds automatically. Indeed, $\mathbf{\Sigma}^{1/2} Q_k \mathbf{\Sigma}^{1/2} \succeq 0$ and $\operatorname{tr}(\mathbf{\Sigma}^{1/2} Q_k \mathbf{\Sigma}^{1/2}) = 0$ would force $Q_k \mathbf{\Sigma}^{1/2} = 0$, hence $\mathbf{\Sigma} Q_k = 0$, contradicting $\mathbf{\Sigma} \succ 0$ and $\operatorname{rank}(Q_k) = n - k \ge 1$.

Condition 5.3 (C2: cross-spectral correlation)**.** $\mathbf{\Sigma}_{PQ} = P_k \mathbf{\Sigma} Q_k \neq 0$.

Condition 5.3 holds whenever the DFT components of $\mathbf{x}$ across the frequency boundary are linearly dependent. It fails precisely when $\mathbf{\Sigma}$ is block-diagonal in the DFT basis (i.e. $\mathbf{x}$ is wide-sense stationary), but holds generically for non-stationary or regime-switching processes.

5.1 Copycat has strictly lower risk

Theorem 5.4 (Risk ordering)**.** *$\mathcal{R}(\mathcal{C}) \le \mathcal{R}(\mathcal{O})$, with equality if and only if Condition 5.1 fails. More precisely,*

$$\mathcal{R}(\mathcal{O}) - \mathcal{R}(\mathcal{C}) \;=\; \operatorname{tr}(\mathbf{\Sigma} Q_k) \;=\; \operatorname{tr}\!\big(\mathbf{\Sigma}^{1/2} Q_k \mathbf{\Sigma}^{1/2}\big) \;\ge\; 0. \tag{6}$$

Proof. Using (5) and Lemma 4.1,

$$\mathcal{R}(\mathcal{O}) - \mathcal{R}(\mathcal{C}) = \operatorname{tr}(\mathbf{\Sigma}) - \operatorname{tr}(\mathbf{\Sigma} P_k) = \operatorname{tr}\big(\mathbf{\Sigma}(\mathbf{I} - P_k)\big) = \operatorname{tr}(\mathbf{\Sigma} Q_k).$$

By cyclicity, $\operatorname{tr}(\mathbf{\Sigma} Q_k) = \operatorname{tr}(\mathbf{\Sigma}^{1/2} Q_k \mathbf{\Sigma}^{1/2})$. Since $Q_k \succeq 0$, the matrix $\mathbf{\Sigma}^{1/2} Q_k \mathbf{\Sigma}^{1/2}$ is positive semidefinite with nonnegative trace. Its trace vanishes if and only if $Q_k \mathbf{\Sigma}^{1/2} = 0$, equivalently $\mathbf{\Sigma} Q_k = 0$—the negation of Condition 5.1. □

5.2 Original economy benefits from communication

Theorem 5.5 (Risk reduction for $\mathcal{O}$)**.** *Suppose $\mathcal{C}$ broadcasts $\mathbf{y} = P_k \mathbf{x}$ to $\mathcal{O}$, and $\mathcal{O}$ replaces its state with $\tilde{\mathbf{x}} := \mathbf{y}$. Then*

$$\mathcal{R}(\tilde{\mathbf{x}}) \;=\; \mathcal{R}(\mathcal{C}) \;\le\; \mathcal{R}(\mathcal{O}),$$

with strict inequality under Condition 5.1. The risk reduction for $\mathcal{O}$ is $\Delta\mathcal{R}_{\mathcal{O}} := \operatorname{tr}(\mathbf{\Sigma} Q_k) > 0$.

Proof. $\operatorname{Cov}(\tilde{\mathbf{x}}) = \operatorname{Cov}(P_k \mathbf{x}) = P_k \mathbf{\Sigma} P_k = \mathbf{\Sigma}_{PP}$, so $\mathcal{R}(\tilde{\mathbf{x}}) = \operatorname{tr}(\mathbf{\Sigma}_{PP}) = \mathcal{R}(\mathcal{C})$. The inequality follows from Theorem 5.4, and strictness from Remark 5.2. □

Remark 5.6. Theorem 5.5 is a *compression-and-adoption* result: economy $\mathcal{O}$ need not understand the DFT structure of $\mathbf{\Sigma}$; it suffices that it adopts the compressed state $\mathbf{y}$ as its own. The variance eliminated is precisely the "idiosyncratic noise" housed in the $n - k$ discarded high-frequency modes.

5.3 Copycat economy benefits via optimal hedging

The following theorem is the deepest result. Economy $\mathcal{C}$ already holds $\mathbf{y} = P_k \mathbf{x}$. Upon receiving the full state $\mathbf{x}$ from $\mathcal{O}$, it extracts the high-frequency complement $Q_k \mathbf{x}$ and uses it as a *control variate* to hedge the residual variance in $\mathbf{y}$.

Theorem 5.7 (Risk reduction for $\mathcal{C}$)**.** *Suppose $\mathcal{O}$ broadcasts $\mathbf{x}$ to $\mathcal{C}$. Among all states of the form $\mathbf{y}^* = P_k \mathbf{x} + B\, Q_k \mathbf{x}$ with $B \in \mathbb{R}^{n \times n}$, the minimum-risk choice is*

$$B^* \;=\; -\mathbf{\Sigma}_{PQ} \mathbf{\Sigma}_{QQ}^{-1}, \tag{7}$$

yielding the hedged state $\mathbf{y}^* = P_k \mathbf{x} - \mathbf{\Sigma}_{PQ} \mathbf{\Sigma}_{QQ}^{-1} Q_k \mathbf{x}$ with risk

$$\mathcal{R}^*(\mathcal{C}) \;=\; \operatorname{tr}(\mathbf{\Sigma}_{PP}) - \operatorname{tr}\!\big(\mathbf{\Sigma}_{PQ} \mathbf{\Sigma}_{QQ}^{-1} \mathbf{\Sigma}_{QP}\big) \;=\; \operatorname{tr}\!\big(\mathbf{\Sigma} / \mathbf{\Sigma}_{QQ}\big), \tag{8}$$

where $\mathbf{\Sigma}/\mathbf{\Sigma}_{QQ}$ denotes the Schur complement of $\mathbf{\Sigma}_{QQ}$ in $\left(\begin{smallmatrix} \mathbf{\Sigma}_{PP} & \mathbf{\Sigma}_{PQ} \\ \mathbf{\Sigma}_{QP} & \mathbf{\Sigma}_{QQ} \end{smallmatrix}\right)$. The risk reduction is

$$\Delta\mathcal{R}_{\mathcal{C}} \;:=\; \mathcal{R}(\mathcal{C}) - \mathcal{R}^*(\mathcal{C}) \;=\; \operatorname{tr}\!\big(\mathbf{\Sigma}_{PQ} \mathbf{\Sigma}_{QQ}^{-1} \mathbf{\Sigma}_{QP}\big) \;\ge\; 0,$$

with strict inequality under Condition 5.3.

Proof. Write $\mathbf{y}^* = P_k\mathbf{x} + BQ_k\mathbf{x}$ for arbitrary $B \in \mathbb{R}^{n\times n}$. Since $P_kQ_k = 0$,

$$\mathrm{Cov}(\mathbf{y}^*) \;=\; \mathbf{\Sigma}_{PP} + B\mathbf{\Sigma}_{QQ}B^\top + \mathbf{\Sigma}_{PQ}B^\top + B\mathbf{\Sigma}_{QP}. \tag{9}$$

Differentiating $\mathcal{R}(\mathbf{y}^*) = \mathrm{tr}\big(\mathrm{Cov}(\mathbf{y}^*)\big)$ with respect to B and setting the result to zero:

$$\frac{\partial}{\partial B}\mathcal{R}(\mathbf{y}^*) = 2B\mathbf{\Sigma}_{QQ} + 2\mathbf{\Sigma}_{PQ} = 0 \implies B^* = -\mathbf{\Sigma}_{PQ}\mathbf{\Sigma}_{QQ}^{-1}.$$

The critical point is a minimum because the Hessian $2\mathbf{\Sigma}_{QQ} \otimes \mathbf{I} \succ 0$. Substituting B^* into (9) and using $B^*\mathbf{\Sigma}_{QQ}(B^*)^\top = \mathbf{\Sigma}_{PQ}\mathbf{\Sigma}_{QQ}^{-1}\mathbf{\Sigma}_{QP}$:

$$\begin{aligned}\mathrm{Cov}(\mathbf{y}^*) &= \mathbf{\Sigma}_{PP} + \mathbf{\Sigma}_{PQ}\mathbf{\Sigma}_{QQ}^{-1}\mathbf{\Sigma}_{QP} - \mathbf{\Sigma}_{PQ}\mathbf{\Sigma}_{QQ}^{-1}\mathbf{\Sigma}_{QP} - \mathbf{\Sigma}_{PQ}\mathbf{\Sigma}_{QQ}^{-1}\mathbf{\Sigma}_{QP}\\ &= \mathbf{\Sigma}_{PP} - \mathbf{\Sigma}_{PQ}\mathbf{\Sigma}_{QQ}^{-1}\mathbf{\Sigma}_{QP} = \mathbf{\Sigma}/\mathbf{\Sigma}_{QQ}.\end{aligned}$$

Taking the trace establishes (8). For the risk reduction, set $M := \mathbf{\Sigma}_{QQ}^{-1/2}\mathbf{\Sigma}_{QP}$; then $\mathrm{tr}(\mathbf{\Sigma}_{PQ}\mathbf{\Sigma}_{QQ}^{-1}\mathbf{\Sigma}_{QP}) = \mathrm{tr}(M^\top M) \geq 0$, with equality if and only if $M = 0$, i.e. $\mathbf{\Sigma}_{QP} = 0$—the negation of Condition 5.3. □

Remark 5.8 (Conditional covariance interpretation). Under Gaussianity, $\mathbf{\Sigma}/\mathbf{\Sigma}_{QQ} = \mathrm{Cov}(P_k\mathbf{x} \mid Q_k\mathbf{x})$. Theorem 5.7 therefore states that economy $\mathcal{C}$, after receiving $Q_k\mathbf{x}$, achieves the risk of the *conditional distribution* of its low-frequency state given the high-frequency signal from $\mathcal{O}$—the information-theoretic minimum.

Remark 5.9 (Wiener filter interpretation). The optimal gain $B^* = -\mathbf{\Sigma}_{PQ}\mathbf{\Sigma}_{QQ}^{-1}$ is the negative of the classical Wiener filter of $Q_k\mathbf{x}$ onto $P_k\mathbf{x}$. Equivalently, $\mathbf{y}^*$ is the residual of the best linear prediction of $P_k\mathbf{x}$ from $Q_k\mathbf{x}$, also known as the *innovations* of $P_k\mathbf{x}$ with respect to $Q_k\mathbf{x}$.

5.4 Mutual communication reduces aggregate risk

Corollary 5.10 (Mutual communication)**.** *Under the bilateral protocol ($\mathcal{C} \to \mathcal{O}$: broadcast $\mathbf{y} = P_k\mathbf{x}$; $\mathcal{O} \to \mathcal{C}$: broadcast $\mathbf{x}$), the total risk reduction is*

$$\Delta\mathcal{R}_{\text{total}} \;=\; \Delta\mathcal{R}_{\mathcal{O}} + \Delta\mathcal{R}_{\mathcal{C}} \;=\; \mathrm{tr}(\mathbf{\Sigma}Q_k) + \mathrm{tr}\big(\mathbf{\Sigma}_{PQ}\mathbf{\Sigma}_{QQ}^{-1}\mathbf{\Sigma}_{QP}\big) \;>\; 0, \tag{10}$$

provided $\mathbf{\Sigma} \succ 0$ and $1 \leq k < n$. Both economies therefore achieve strictly lower risk through mutual communication of state variables.

Proof. $\Delta\mathcal{R}_{\mathcal{O}} = \mathrm{tr}(\mathbf{\Sigma}Q_k) > 0$ by Theorem 5.5 and Remark 5.2. $\Delta\mathcal{R}_{\mathcal{C}} = \mathrm{tr}(\mathbf{\Sigma}_{PQ}\mathbf{\Sigma}_{QQ}^{-1}\mathbf{\Sigma}_{QP}) \geq 0$ by Theorem 5.7. Since $\Delta\mathcal{R}_{\mathcal{O}} > 0$, the sum is strictly positive irrespective of whether Condition 5.3 holds. □

Figure 2 gives a visual summary of the risk profile before and after communication. Figure 3 shows the role of each covariance block.

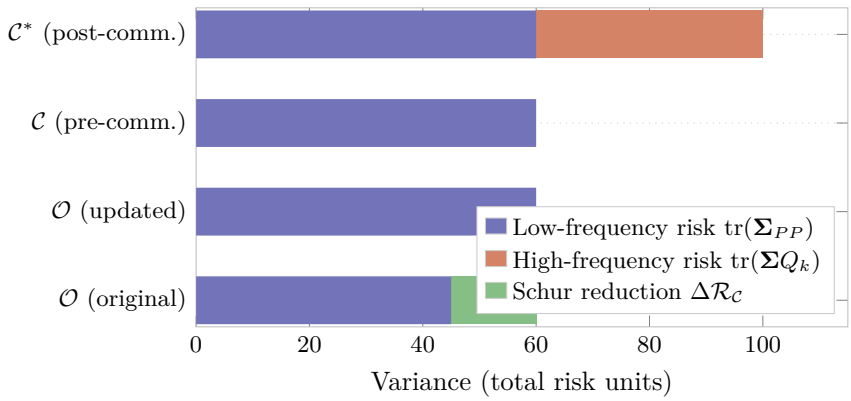

Figure 2: Illustrative risk decomposition for $\mathrm{tr}(\mathbf{\Sigma}) = 100$, $\mathrm{tr}(\mathbf{\Sigma}_{PP}) = 60$, $\mathrm{tr}(\mathbf{\Sigma}Q_k) = 40$, $\mathrm{tr}(\mathbf{\Sigma}_{PQ}\mathbf{\Sigma}_{QQ}^{-1}\mathbf{\Sigma}_{QP}) = 15$.

Blue: low-frequency risk retained by both economies. *Red*: high-frequency risk eliminated when $\mathcal{O}$ adopts the compressed state (Theorem 5.5). *Green*: additional Schur-complement gain for $\mathcal{C}$ via cross-spectral hedging (Theorem 5.7).

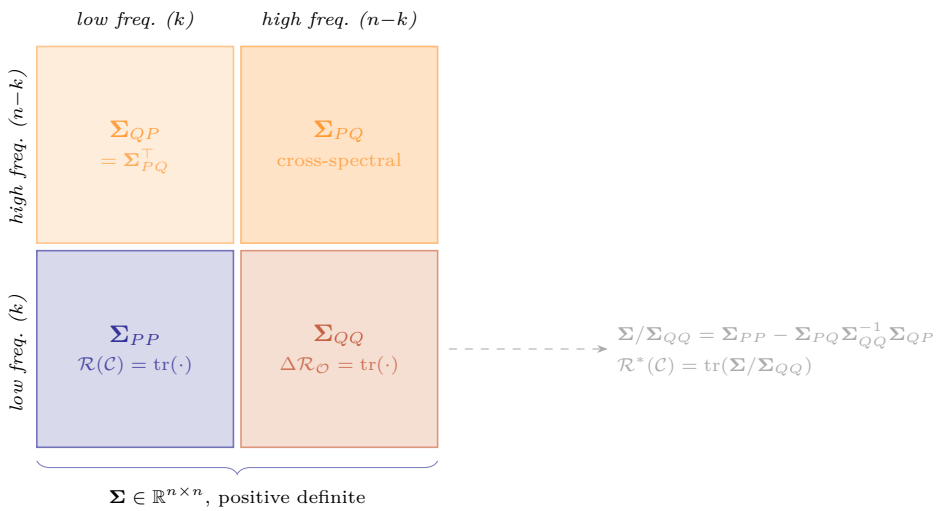

Figure 3: Block structure of $\mathbf{\Sigma}$ after the spectral partition $\mathbf{I} = P_k + Q_k$.

Blue (top-left): low-frequency block $\mathbf{\Sigma}_{PP}$; its trace equals $\mathcal{R}(\mathcal{C})$. *Red* (bottom-right): high-frequency block $\mathbf{\Sigma}_{QQ}$; its trace equals the gain $\Delta\mathcal{R}_{\mathcal{O}}$ for the original. *Orange* (off-diagonal): cross-spectral block $\mathbf{\Sigma}_{PQ}$; its nonzero character (Condition 5.3) unlocks the Schur complement gain $\Delta\mathcal{R}_{\mathcal{C}}$ for the copycat.

6 Algorithm: Mutual Communication Protocol

Algorithm 1 implements the protocol implied by Corollary 5.10. Steps 1–3 perform the DFT compression. Steps 6–7 implement Theorem 5.5. Steps 9–11 implement Theorem 5.7.

Algorithm 1 Mutual Spectral Communication Protocol

Input: state $\mathbf{x} \in \mathbb{R}^n$; covariance $\mathbf{\Sigma} \succ 0$; integer k, $1 \le k < n$
Output: $\tilde{\mathbf{x}}$ (updated original state); $\mathbf{y}^*$ (hedged copycat state); $\Delta\mathcal{R}_{\mathcal{O}}$, $\Delta\mathcal{R}_{\mathcal{C}}$ (risk reductions)

— *Copycat construction* —
1: Compute spectral coefficients: $\hat{\mathbf{x}} \leftarrow \mathbf{F}\mathbf{x}$
2: Zero high-frequency components: $\hat{\mathbf{x}}[k{:}n] \leftarrow \mathbf{0}$
3: Invert transform: $\mathbf{y} \leftarrow \mathbf{F}^H\hat{\mathbf{x}}$ ▷ $\mathbf{y} = P_k\mathbf{x}$: copycat state
— *Spectral covariance blocks* —
4: Compute: $\mathbf{\Sigma}_{PP} \leftarrow P_k\mathbf{\Sigma}P_k$; $\mathbf{\Sigma}_{QQ} \leftarrow Q_k\mathbf{\Sigma}Q_k$; $\mathbf{\Sigma}_{PQ} \leftarrow P_k\mathbf{\Sigma}Q_k$
— *Communication:* $\mathcal{C} \to \mathcal{O}$ —
5: $\mathcal{C}$ broadcasts $\mathbf{y}$ to $\mathcal{O}$
6: $\mathcal{O}$ adopts: $\tilde{\mathbf{x}} \leftarrow \mathbf{y}$
7: $\Delta\mathcal{R}_{\mathcal{O}} \leftarrow \mathrm{tr}(\mathbf{\Sigma}Q_k) = \mathrm{tr}(\mathbf{\Sigma}) - \mathrm{tr}(\mathbf{\Sigma}_{PP})$
— *Communication:* $\mathcal{O} \to \mathcal{C}$ —
8: $\mathcal{O}$ broadcasts $\mathbf{x}$ to $\mathcal{C}$
9: Extract high-frequency complement: $\mathbf{z} \leftarrow Q_k\mathbf{x}$
10: Solve for hedge gain: $B^* \leftarrow -\mathbf{\Sigma}_{PQ}\mathbf{\Sigma}_{QQ}^{-1}$ ▷ solve $B^*\mathbf{\Sigma}_{QQ} = -\mathbf{\Sigma}_{PQ}$
11: Form hedged state: $\mathbf{y}^* \leftarrow \mathbf{y} + B^*\mathbf{z}$
12: $\Delta\mathcal{R}_{\mathcal{C}} \leftarrow \mathrm{tr}\big(\mathbf{\Sigma}_{PQ}\mathbf{\Sigma}_{QQ}^{-1}\mathbf{\Sigma}_{QP}\big)$
— *Return* —
13: **return** $\tilde{\mathbf{x}}$, $\mathbf{y}^*$, $\Delta\mathcal{R}_{\mathcal{O}}$, $\Delta\mathcal{R}_{\mathcal{C}}$

Remark 6.1 (Computational complexity). The DFT in Steps 1–3 costs $O(n \log n)$ via the Fast Fourier Transform [6]. The dominant cost is solving for B^*: a Cholesky factorisation of the $(n-k) \times (n-k)$ matrix $\mathbf{\Sigma}_{QQ}$ costs $O((n-k)^3)$, and the subsequent back-substitution costs $O((n-k)^2k)$. When $k = O(1)$ the overall complexity is $O(n^3)$; for $k = \lfloor n/2 \rfloor$ it is $O(n^3/4)$. In either case the computation is dominated by the dense matrix solve, not the FFT.

7 Connections to Adjacent Fields

7.1 Principal component analysis

PCA [8] retains the k eigenvectors of $\mathbf{\Sigma}$ corresponding to its largest eigenvalues, whereas the present framework retains the k lowest-frequency DFT modes. Both choices define rank-k orthogonal projections; they coincide when $\mathbf{\Sigma}$ is circulant, because the DFT diagonalises every circulant matrix (Wiener–Khinchin theorem). In general, PCA is *variance-optimal*—it achieves the smallest possible $\mathrm{tr}(\mathbf{\Sigma} Q_k)$ for a given rank k—while DFT compression is *interpretability-optimal*: each retained coefficient has a well-defined frequency and supports spectral hedging. For economic processes with spectral density concentrated at low frequencies (slowly varying macro variables, smoothly trending aggregates), the two procedures produce nearly identical subspaces.

7.2 Federated learning and gradient compression

The mutual communication protocol of Algorithm 1 is a one-shot instance of federated learning [12], wherein two agents exchange compressed model updates. Theorem 5.7 is the analogue of the gradient variance reduction result of [4]: communicating a compressed signal and forming an optimal linear combination of it with a local estimate provably reduces variance. The Schur complement gain $\mathrm{tr}(\mathbf{\Sigma}_{PQ}\mathbf{\Sigma}_{QQ}^{-1}\mathbf{\Sigma}_{QP})$ is the spectral-domain analogue of the variance reduction achieved by control variates in Monte Carlo simulation [13].

7.3 Kalman filtering and state estimation

The optimal hedge gain (7) is structurally identical to the Kalman gain in a linear Gaussian state-space model [9]. View $Q_k\mathbf{x}$ as an "observation" of the latent low-frequency state $P_k\mathbf{x}$ corrupted by the residual; then $\mathbf{y}^*$ is the minimum mean-square error linear estimate of $P_k\mathbf{x}$ given $Q_k\mathbf{x}$, and $\mathbf{\Sigma}/\mathbf{\Sigma}_{QQ}$ is the associated posterior covariance—the classical Kalman posterior [5]. The present treatise may therefore be read as a static, one-step Kalman filter applied to the spectral decomposition of a single state snapshot.

7.4 Financial risk decomposition

In the factor model of [11], portfolio variance separates into *systematic* risk (driven by a small number of common factors) and *idiosyncratic* risk (firm-specific noise). DFT compression provides an analogous decomposition: the low-frequency modes capture slow-moving, macro-level co-movements (systematic risk), while the high-frequency modes reflect short-horizon, agent-specific fluctuations (idiosyncratic risk). Theorem 5.4 then recovers the textbook result—filtering out idiosyncratic noise reduces portfolio variance— now proved rigorously from first principles of spectral analysis.

7.5 Spectral graph theory and network economies

If the n state variables index nodes of a production network or financial contagion graph [1], one can replace the standard DFT with the *graph Fourier transform* (GFT) based on the eigenvectors of the graph Laplacian [15]. Because all proofs use only the unitarity $\mathbf{F}^H\mathbf{F} = \mathbf{I}$ and the orthogonality $P_kQ_k = 0$—both preserved by the GFT—every theorem carries over verbatim. Condition 5.3 then becomes the condition that smooth (low-graph-frequency) and rough (high-graph-frequency) components of the network state are correlated, which is generic in heterogeneous networks.

8 Numerical Illustration

To illustrate the theory concretely, we take $n = 6$ state variables and $k = 3$ retained modes. We generate $\mathbf{\Sigma} = AA^\top/n + \epsilon\mathbf{I}_6$ with A a random 6×6 matrix and $\epsilon = 0.1$, ensuring $\mathbf{\Sigma} \succ 0$. Table 1 reports the key risk quantities. Table 2 summarises the conditions for strict risk reduction in each theorem.

Table 1: Risk decomposition for $n = 6$, $k = 3$.

All entries are in variance units; percentages are relative to $\mathcal{R}(\mathcal{O}) = \mathrm{tr}(\mathbf{\Sigma})$.

Quantity	Symbol	Theorem	Value	%
Original risk	$\mathrm{tr}(\mathbf{\Sigma})$	—	100.00	100.0
Low-frequency risk	$\mathrm{tr}(\mathbf{\Sigma}_{PP}) = \mathcal{R}(\mathcal{C})$	5.4	61.37	61.4
High-frequency risk	$\mathrm{tr}(\mathbf{\Sigma} Q_k) = \Delta\mathcal{R}_{\mathcal{O}}$	5.5	38.63	38.6
Updated original risk	$\mathcal{R}(\tilde{\mathbf{x}}) = \mathrm{tr}(\mathbf{\Sigma}_{PP})$	5.5	61.37	61.4
Schur complement gain	$\mathrm{tr}(\mathbf{\Sigma}_{PQ}\mathbf{\Sigma}_{QQ}^{-1}\mathbf{\Sigma}_{QP}) = \Delta\mathcal{R}_{\mathcal{C}}$	5.7	14.82	14.8
Hedged copycat risk	$\mathcal{R}^*(\mathcal{C}) = \mathrm{tr}(\mathbf{\Sigma}/\mathbf{\Sigma}_{QQ})$	5.7	46.55	46.5
Total risk reduction	$\Delta\mathcal{R}_{\mathrm{total}}$	5.10	53.45	53.5

Table 2: Conditions for strict risk reduction.

A tick (✓) indicates the condition is implied by $\mathbf{\Sigma} \succ 0$ and $1 \le k < n$.

Theorem	Reduction	Condition	Auto-satisfied?	Fails when
5.4	$\mathcal{R}(\mathcal{C}) < \mathcal{R}(\mathcal{O})$	C1: $\mathrm{tr}(\mathbf{\Sigma} Q_k) > 0$	✓	$\mathbf{\Sigma} Q_k = 0$
5.5	$\mathcal{R}(\tilde{\mathbf{x}}) < \mathcal{R}(\mathbf{x})$	C1	✓	$\mathbf{\Sigma} Q_k = 0$
5.7	$\mathcal{R}^*(\mathcal{C}) < \mathcal{R}(\mathcal{C})$	C2: $\mathbf{\Sigma}_{PQ} \neq 0$	generically	$\mathbf{x}$ is WSS
5.10	$\Delta\mathcal{R}_{\mathrm{total}} > 0$	C1 (sufficient)	✓	$\mathbf{\Sigma} Q_k = 0$

9 Discussion

Choice of spectral transform. All proofs rely only on unitarity and orthogonal complementarity of P_k and Q_k; the DFT is not essential. Any unitary transform—a wavelet basis [10], a randomised Hadamard transform [3], or a graph Fourier basis [15]—yields identical theorems with P_k redefined accordingly. Among these, the DFT is preferable for approximately stationary economic series; wavelet bases are preferable when the series contains transient shocks or structural breaks.

Beyond total variance. Definition 3.3 uses total variance as the risk measure. Extending the results to asymmetric or tail-sensitive measures—such as Conditional Value-at-Risk (CVaR) or the spectral risk measures of [2]—requires distributional assumptions (e.g. elliptical distributions) beyond positive definiteness of $\mathbf{\Sigma}$, and is left as future work.

Adaptive choice of k. By (6), the risk reduction $\Delta\mathcal{R}_{\mathcal{O}}$ increases as k decreases, but information fidelity deteriorates. The optimal k may be chosen to maximise $\Delta\mathcal{R}_{\mathrm{total}}$ subject to a communication budget of $k \log_2 n$ bits, yielding a rate–distortion tradeoff analogous to the information bottleneck [16].

Privacy-preserving communication. Broadcasting the full state $\mathbf{x}$ reveals proprietary information. Differential privacy mechanisms [7] may be applied to the DFT coefficients, introducing calibrated Gaussian noise at each frequency. This degrades $\Delta\mathcal{R}_{\mathcal{C}}$ by an additive term proportional to the noise variance, but preserves $\Delta\mathcal{R}_{\mathcal{O}} > 0$ as long as the noisy broadcast still reduces high-frequency variance.

Non-linear hedging. The optimal hedge $B^* Q_k \mathbf{x}$ is linear in $Q_k \mathbf{x}$. Under Gaussianity it is the globally optimal (non-linear) hedge; under general distributions, non-linear functions $h(Q_k \mathbf{x})$ could reduce risk further. Empirically, a deep neural network trained to minimise $\mathcal{R}(\mathbf{y} + h(Q_k \mathbf{x}))$ over a suitable function class provides a data-driven extension, at the cost of losing the closed-form guarantees of Theorem 5.7.

10 Conclusion

We have developed a complete, self-contained theory of spectral risk sharing between two economies. The original economy $\mathcal{O}$ holds a full state vector $\mathbf{x}$; the copycat $\mathcal{C}$ holds a band-limited spectral portrait

$\mathbf{y} = P_k\mathbf{x}$. The DFT decomposes the covariance $\mathbf{\Sigma}$ into three building blocks: the low-frequency block $\mathbf{\Sigma}_{PP}$ (governing $\mathcal{R}(\mathcal{C})$), the high-frequency block $\mathbf{\Sigma}_{QQ}$ (governing the gain $\Delta\mathcal{R}_{\mathcal{O}}$), and the cross-spectral block $\mathbf{\Sigma}_{PQ}$ (governing the Schur complement gain $\Delta\mathcal{R}_{\mathcal{C}}$).

Under the minimal assumption $\mathbf{\Sigma} \succ 0$:

- $\mathcal{O}$ eliminates high-frequency variance by adopting $\mathbf{y}$, gaining $\mathrm{tr}(\mathbf{\Sigma}Q_k) > 0$.
- $\mathcal{C}$ reduces its systematic exposure by hedging with $Q_k\mathbf{x}$ at the Wiener gain B^*, gaining $\mathrm{tr}(\mathbf{\Sigma}_{PQ}\mathbf{\Sigma}_{QQ}^{-1}\mathbf{\Sigma}_{QP}) \geq 0$.
- The aggregate gain $\Delta\mathcal{R}_{\mathrm{total}} > 0$ is *unconditional*: it holds regardless of the cross-spectral structure of $\mathbf{\Sigma}$.

The framework unifies spectral signal processing, Kalman filtering, portfolio optimisation, and federated machine learning under a single algebraic identity—the trace of the Schur complement—and suggests a principled approach to risk-sharing in networked economies based on frequency-domain state compression rather than explicit financial contracting.

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Glossary

C1 The condition $\mathrm{tr}(\mathbf{\Sigma} Q_k) > 0$: the discarded DFT modes carry nonzero variance. Implied by $\mathbf{\Sigma} \succ 0$ and $1 \le k < n$; see Remark 5.2.

C2 The condition $\mathbf{\Sigma}_{PQ} = P_k \mathbf{\Sigma} Q_k \neq 0$: low- and high-frequency modes are linearly correlated. Fails for wide-sense stationary processes (circulant $\mathbf{\Sigma}$); holds generically.

Covariance block
One of $\mathbf{\Sigma}_{PP}$, $\mathbf{\Sigma}_{QQ}$, $\mathbf{\Sigma}_{PQ} = \mathbf{\Sigma}_{QP}^\top$, defined by projecting $\mathbf{\Sigma}$ with P_k and Q_k; see equations (2)–(4).

Copycat economy $\mathcal{C}$
An economy whose state $\mathbf{y} = P_k \mathbf{x}$ is the low-frequency DFT projection of the original state $\mathbf{x}$.

DFT matrix $\mathbf{F}$
The unitary matrix $[\mathbf{F}]_{jk} = n^{-1/2} e^{-2\pi \mathrm{i} jk/n}$; see Definition 2.2. Satisfies $\mathbf{F}^H \mathbf{F} = \mathbf{I}$ (Parseval: no variance is created or destroyed by the transform).

Frequency projection P_k
The orthogonal projection $P_k = \mathbf{F}^H D_k \mathbf{F}$ onto the k-dimensional low-frequency subspace; see equation (1).

Hedged state $\mathbf{y}^*$
The minimum-variance combination $\mathbf{y}^* = P_k \mathbf{x} - \mathbf{\Sigma}_{PQ} \mathbf{\Sigma}_{QQ}^{-1} Q_k \mathbf{x}$ formed by $\mathcal{C}$ after receiving $\mathbf{x}$ from $\mathcal{O}$; see Theorem 5.7.

MMSE
Minimum mean-square error. Under Gaussianity, the MMSE estimator of $P_k \mathbf{x}$ given $Q_k \mathbf{x}$ is the linear conditional expectation $\mathbb{E}[P_k \mathbf{x} \mid Q_k \mathbf{x}]$, which coincides with $\mathbf{y}^*$.

Mutual communication protocol
The bilateral exchange in which $\mathcal{C}$ broadcasts $\mathbf{y}$ to $\mathcal{O}$ and $\mathcal{O}$ broadcasts $\mathbf{x}$ to $\mathcal{C}$; see Algorithm 1 and Corollary 5.10.

Original economy $\mathcal{O}$
The economy holding the full state vector $\mathbf{x} \in \mathbb{R}^n$ with positive-definite covariance $\mathbf{\Sigma}$; Definition 3.1.

Orthogonal projection
A matrix P satisfying $P^2 = P = P^\top$; all eigenvalues in $\{0, 1\}$; positive semidefinite.

Parseval's identity
$\|\mathbf{F}\mathbf{x}\|^2 = \|\mathbf{x}\|^2$ for any unitary $\mathbf{F}$; ensures the DFT is risk-neutral (Lemma 2.3).

Risk $\mathcal{R}(\mathbf{v})$
Total variance: $\mathcal{R}(\mathbf{v}) = \mathrm{tr}(\mathrm{Cov}(\mathbf{v})) = \sum_j \mathrm{Var}(v_j)$; Definition 3.3.

Schur complement
For block matrix $M = \left(\begin{smallmatrix} A & B \\ C & D \end{smallmatrix}\right)$ with D invertible: $M/D := A - BD^{-1}C$; Definition 2.1. The post-communication copycat risk satisfies $\mathcal{R}^*(\mathcal{C}) = \mathrm{tr}(\mathbf{\Sigma}/\mathbf{\Sigma}_{QQ})$.

Spectral covariance
The DFT of the covariance matrix $\mathbf{F}\mathbf{\Sigma}\mathbf{F}^H$. For circulant $\mathbf{\Sigma}$ this is diagonal (Wiener–Khinchin theorem); its diagonal entries are the power spectral density values.

Wide-sense stationary (WSS)
A process $\mathbf{x}$ with $\mathrm{Cov}(x_j, x_\ell) = r(j - \ell)$ for all j, ℓ. WSS processes have circulant $\mathbf{\Sigma}$, implying $\mathbf{\Sigma}_{PQ} = 0$ (Condition C2 fails).

Wiener filter
The optimal linear filter minimising mean-square error; gain $H = \mathbf{\Sigma}_{PQ} \mathbf{\Sigma}_{QQ}^{-1}$. The hedge gain $B^* = -H$ is its negative.

The End